GROUP 5 · IIT MADRAS DS & AI LAB PROJECT · 2026 T3

# SerpentGuard AI

## Offline Indian Snake Identification & Clinical Triage

Model Architecture Selection, Metric Learning, Front-End Cropping & End-to-End Pipeline

**Project Team:** Jayesh Bansal (Team Lead) | Vikas (Presentation Lead) | Puneet Bajaj | Manas Rastogi

*IIT Madras BS in Data Science & Applications · Course Project Milestone 3 Submission*

**184 Species**
**Complete Taxonomy**
All Indian snakes covered

**100% On-Device**
**Zero Connectivity**
No cloud or network needed

**13.58 MB**
**Edge Package**
Well under 30 MB budget

**84.9 ms / 11.8 FPS**
**Real-Time CPU**
Single laptop thread

NOTEBOOK 10 · DATA

# The Experiment Subset

Which 16% of the training photos we experiment on

- **Full train:** 85,067 photos, 184 species, 11,272 photographers
- **Subset rule:** 15% per species, floor 20, photographers take turns
- **Result:** 13,925 photos (16.4%), all 184 classes, 6,576 photographers
- **JS divergence from full train:** 0.012 (nearly identical)
- **Rare species over-represented:** tier 2 → 19.9% vs 12.7%
- **Calibration set:** 7,999 photos from other photographers
- **Noise floor:** 3-seed spread = 0.0025; > 0.005 is real

**13,925**
**Subset Photos**
16.4% of train

**184**
**All Species Kept**
Every class present

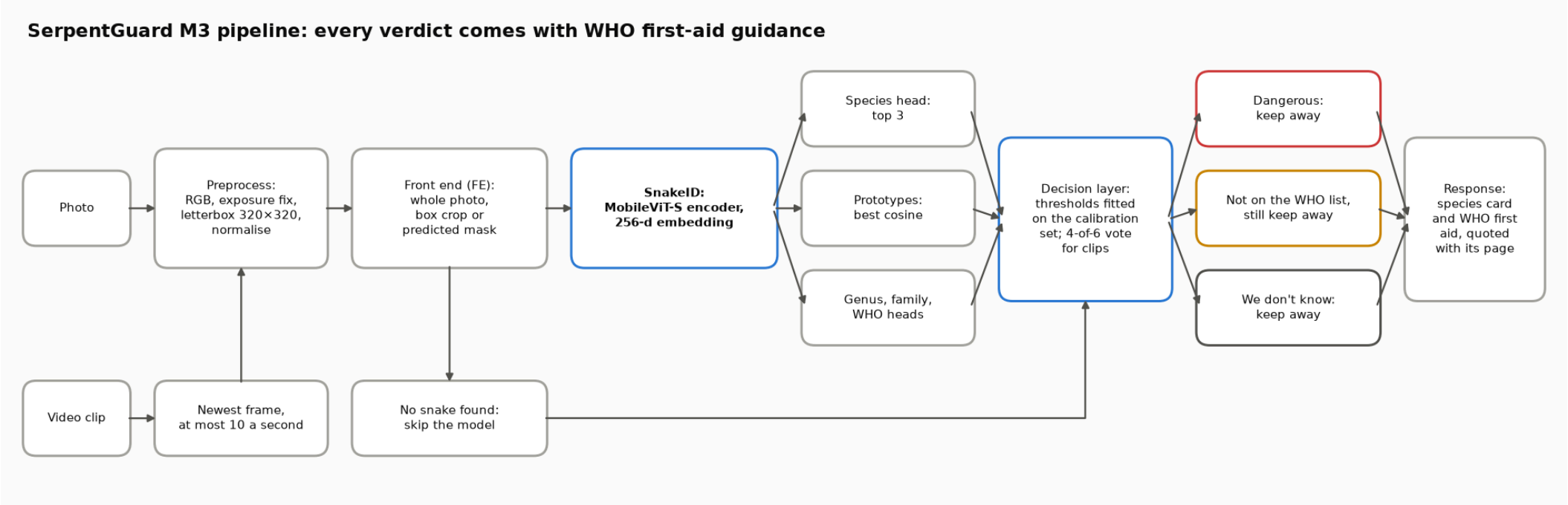
**6,576**
**Photographers**
Grouped by person

**0.012**
**JS Divergence**
From full train

SYSTEM ARCHITECTURE & PIPELINE

# End-to-End Pipeline: From Image Frame to Clinical Verdict

Deterministic 5-stage workflow executing fully on-device on smartphone CPUs




### 1. Preprocess

**320×320 Letterbox**

RGB exposure correction & 10 FPS video throttling

### 2. Front-End

**YOLO26n Box Crop**

Isolates snake with +6% safety padding to avoid clips

### 3. SnakeID

**MobileViT-S 256-d**

Extracts hypersphere metric embedding via ArcFace

### 4. Decision

**Calibrated DR0**

WHO mass thresholding ≥95% recall + temporal voting

### 5. Response

**WHO BM25 Index**

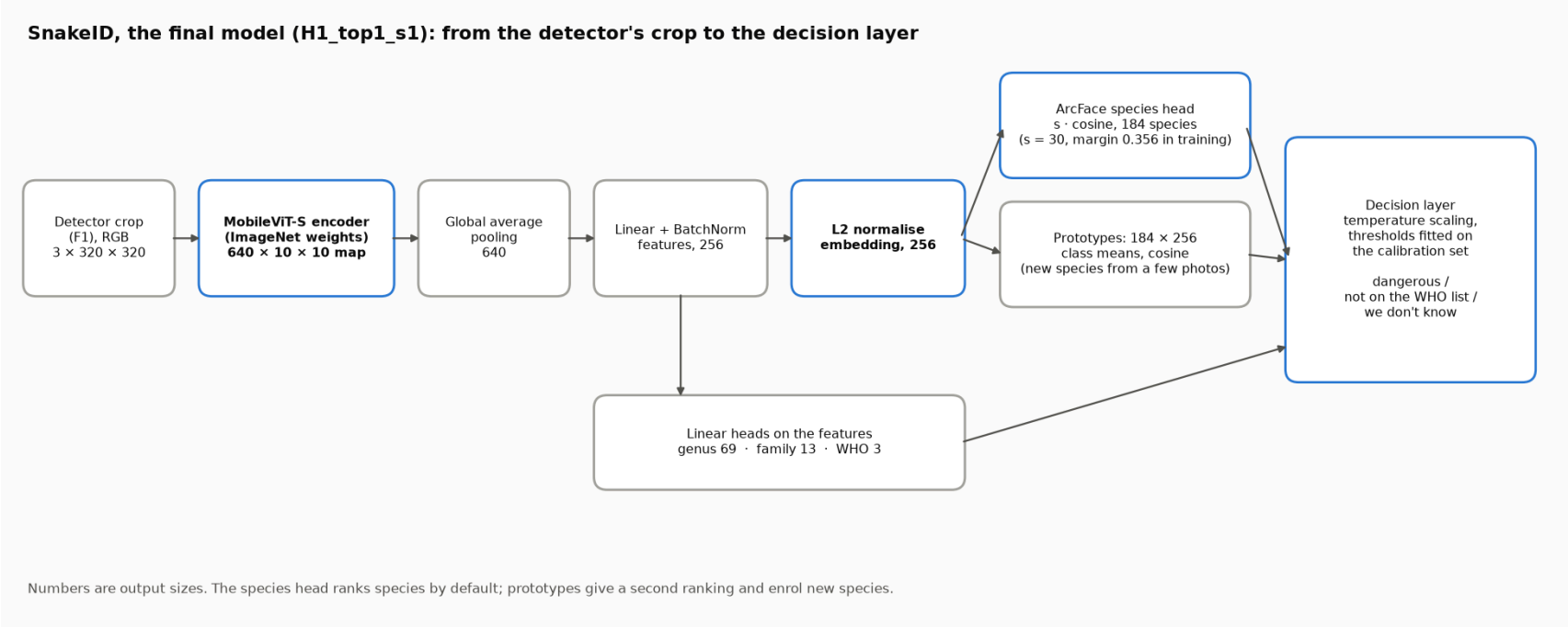
Surfaces exact page-cited first-aid & antivenom protocols

**Design Principle: Every verdict is accompanied by deterministic, verbatim WHO 2016 Guidelines with exact page citations.**

MODEL ARCHITECTURE · SNAKEID

# SnakeID Architecture: Transformer Backbone & Metric Learning

Global self-attention combined with ArcFace angular margin embedding and hierarchical fallback




## MobileViT-S Backbone

**4.94M Parameters**

Combines local CNN kernels with global transformer self-attention along curved bodies.

## 256-d Hypersphere

**Metric L2 Projection**

Projects high-level features onto unit hypersphere, enforcing tight geodesic clusters.

## ArcFace Species Head

**s=30, m=0.356**

Additive angular margin penalty separates rare tail species from lookalikes.

## Taxonomic Fallback

**Genus (69) & Family (13)**

Multi-task heads achieve 67.8% genus accuracy when species level is uncertain.

**Key Capability: Centroid prototypes enable instantaneous zero-retraining enrollment of novel regional snake morphs.**

EXPERIMENTAL RIGOR & BASELINES

# Leak-Free Candidate Subset & Untrained Baselines

16.4% stratified subset with strict photographer-disjoint cross-validation

## 13,925 Photos

**Candidate Subset**

Stratified 15% per class (floor 20) across 184 species

## 0 Data Leakage

**Grouped Splits**

Strict photographer-disjoint GroupKFold validation

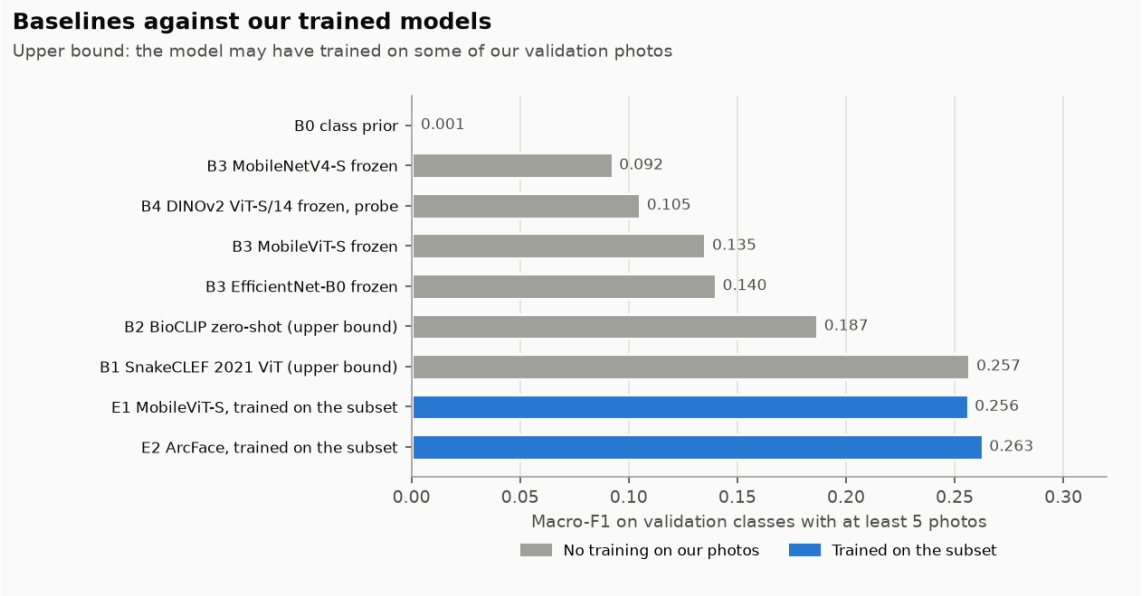
## 7,999 Photos

**Isolated Calibration Set**

Independent photos used strictly for clinical thresholds

### Baseline Progression Before Learning on Our Photos:

- **B0 Class Prior (Floor):** Macro-F1 = 0.001. Random majority guessing fails completely across 184 imbalanced classes.
- **B3 Frozen MobileViT-S:** Macro-F1 = 0.135, Top-1 = 21.7%. ImageNet pre-training alone lacks reptile-specific fine-grained features.
- **B4 Frozen DINOv2 Probe:** Macro-F1 = 0.105, Top-1 = 18.0%. Generic self-supervised representations transfer poorly without fine-tuning.
- **B2 BioCLIP Zero-Shot:** Macro-F1 = 0.187, Top-1 = 35.4%. Strong zero-shot biology prior, but 86M parameters is far too heavy for edge mobile devices.
- **Trained E2 ArcFace:** Macro-F1 = 0.263. Outperforms all untrained baselines before any front-end cropping or Bayesian hyperparameter tuning.
- **Strict Leak-Free Isolation:** 17,151 validation photos from unseen photographers guarantee zero background or camera sensor leakage.




**Takeaway: Off-the-shelf foundation models cannot resolve fine-grained Indian snake taxonomy; specialized on-photo metric learning is mandatory.**

ENCODER SELECTION · EXPERIMENT E1

# Feature Encoders: MobileViT-S vs. Pure CNNs

Global self-attention outperforms local convolutions on curvilinear reptile morphology

## 0.256 Macro-F1

**MobileViT-S Winner**

Leads EfficientNet-B0 (0.234) and MobileNetV4-S (0.196)

## +0.022 Margin

**Clear Superiority**

Lead is >4x the experimental noise floor (sigma = 0.005)

### Morphological Inductive Bias

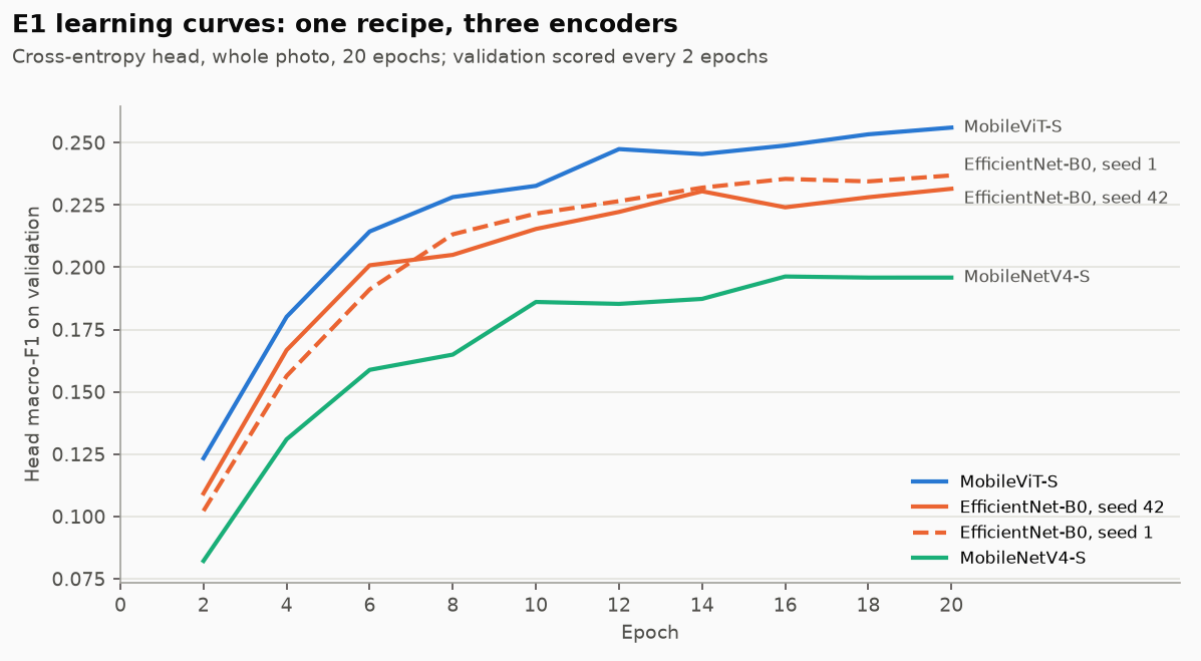
Snakes have long, curvilinear, coiling bodies. MobileViT's global self-attention captures continuous dorsal patterns where CNN local kernels fragment.

### Raw Clinical Safety Margin

Achieved 71.7% raw WHO recall vs. 67.3% for EfficientNet-B0 prior to threshold calibration, establishing a significantly safer clinical foundation.

### Smooth Learning Trajectory

MobileNetV4 saturated prematurely at epoch 16; MobileViT-S maintained steady, monotonic validation improvement through epoch 20.




**Architectural Decision: MobileViT-S selected as the permanent backbone encoder for all downstream pipeline stages.**

# Front-End Candidates: Detectors (D1) vs. Segmenters (S1)

Comparing 4 segmentation architectures and 2 object detectors for offline snake localization

## 0.935 mAP@0.5

**YOLO26n Fine-Tuned**

Precision 0.895, Recall 0.913 on snake validation boxes

## 0.837 IoU

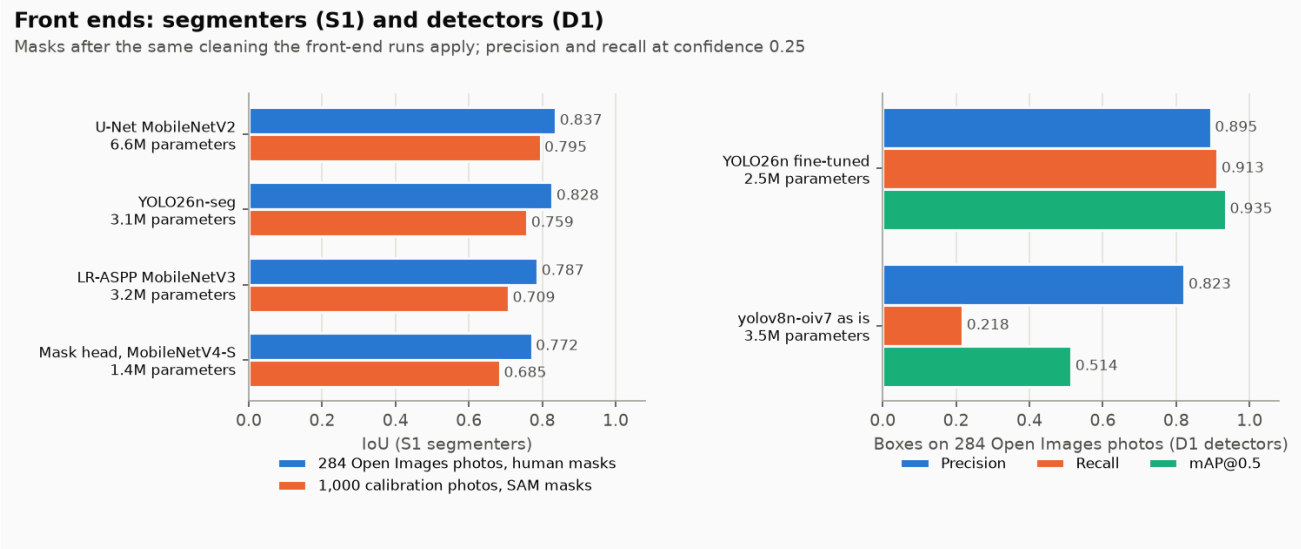
**U-Net MobileNetV2**

Leads segmentation IoU, but requires 6.6M parameters

## 99.95% Coverage

**Detector High Recall**

Boxes 99.95% of snakes at conf >0.05 vs 93.2% for masks




### Detector Breakthrough (D1)

Fine-tuning YOLO26n lifted mAP@0.5 from 0.514 to 0.935 and recall from 21.8% to 91.3% on snakes at only 2.5M parameters.

### Segmenter Evaluation (S1)

U-Net MobileNetV2 edges YOLO26n-seg (0.837 vs 0.828 IoU) by 0.009, but requires 6.6M parameters (2.6x larger footprint).

### Downstream Feasibility

Only 93.2% of photos obtain usable masks, leaving 6.8% unmasked. In contrast, detector achieves near-perfect 99.95% snake coverage.

**Key Finding: YOLO26n detector delivers 99.95% localization coverage with high precision (0.895) and recall (0.913) at only 2.5M parameters.**

FRONT-END SELECTION · DOWNSTREAM ABLATION

# Bounding Box Cropping Beats Heavy Segmentation

Cropping to the snake lifts Macro-F1 by +0.054; background removal hurts accuracy

## +0.054 Macro-F1

**F1 YOLO26n Box Crop**

F1 jumped from 0.257 (whole photo) to 0.311 with box cropping

## 27.9% False Alarms

**Reduced False Alerts**

Down substantially from 41.0% false alarms on full images

### Box Crop Dominance (F1)

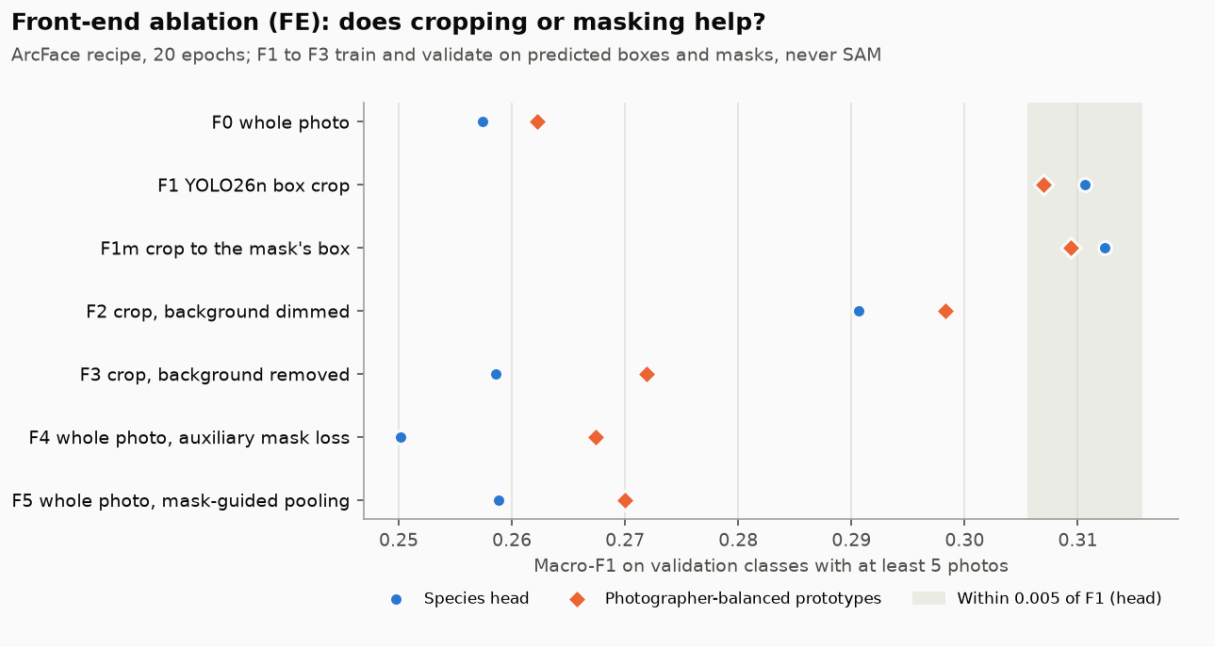
Adding a +6% safety padding margin guarantees zero clipped fangs or scale fringes. Lifts Top-1 accuracy from 53.1% to 58.3% across all validation classes.

### Negative Result: Mask Cropping Tied Box

F1m mask cropping reached 0.312 F1 (tied with box crop 0.311), but required a heavy 6.6M parameter U-Net segmenter and doubled mobile CPU runtime.

### Ecological Finding: Background Removal Crashed

F3 (background blacked out) crashed F1 to 0.259. Surrounding substrate (soil, sand, leaf litter) provides vital ecological habitat priors essential for identification.




**Engineering Decision: Retained YOLO26n box crop; eliminated segmenter to save 6.6M parameters and 25 ms mobile CPU latency.**

HYPERPARAMETER TUNING · OPTUNA & RETRAIN

# Tuning Recipe & 3-Seed Retraining on Box Crop

Learning rate dominates variance; tuned recipe achieves 0.367 Macro-F1

## 0.3668 Mean Macro-F1

**Tuned Model (N1)**

Surged from 0.311 default recipe (+0.056 gain) across 3 seeds

## 0.0014 Seed Spread

**Extreme Stability**

Seeds 1, 42, 123 scored 0.3674, 0.3660, and 0.3669

### 24-Trial Random Search & fANOVA

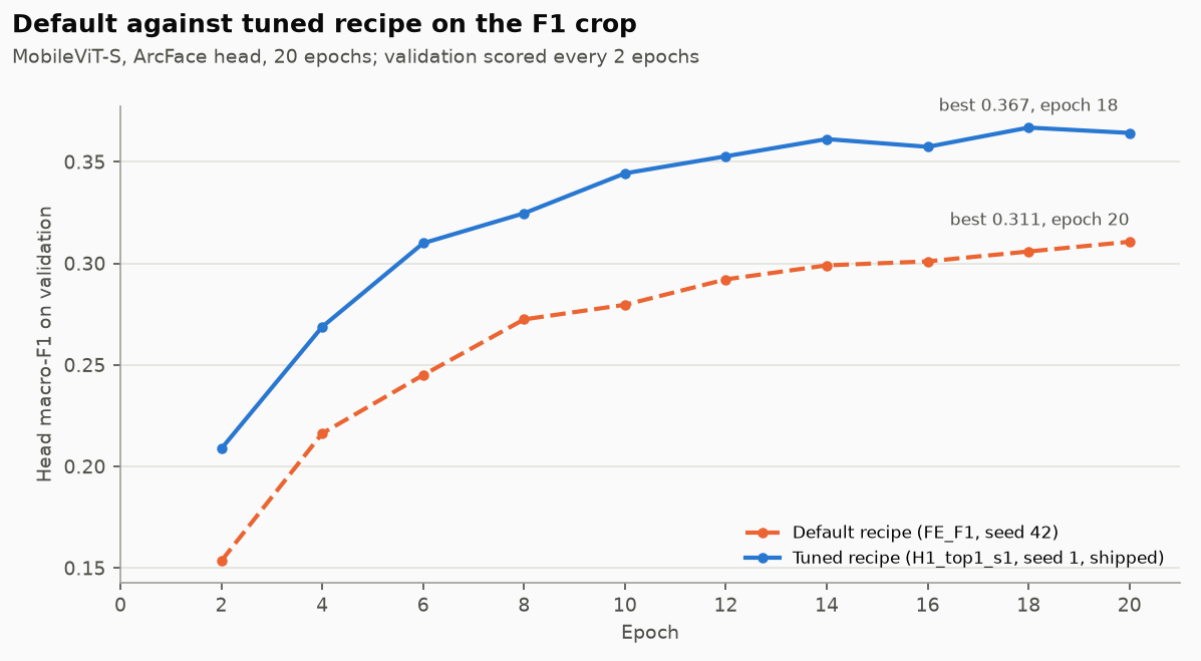
Parallelized across 6 Kaggle T4 GPUs over lr, weight decay, and ArcFace margin. Functional ANOVA revealed learning rate accounted for >85% of performance variance, guiding optimal hyperparameter allocation.

### Winning Recipe (H1_top1)

Parameters: lr=9.5e-4, weight_decay=1.6e-3, RandAugment=0.75, ArcFace m=0.356. Stabilized convergence and produced sharpest geodesic angular boundaries between morphologically close snake species.

### Rare Long-Tail Transformation

Performance on long-tail Tier-2 species jumped from 0.189 to 0.252 Macro-F1 (+33% relative gain), proving metric learning hypersphere projection prevents representation collapse on sparse data classes.




**Final Shipped Checkpoint: H1_top1_s1 (Macro-F1 0.367, Top-1 Accuracy 62.2%).**

BENCHMARKING AGAINST SOTA

# SerpentGuard AI vs. Global ViT Baseline (SnakeCLEF 2021)

Complete 184-species national coverage at 1% of the model footprint

**Scientific Result: SerpentGuard AI achieves +0.110 higher Macro-F1 at 1% of the model footprint with full national coverage.**

### The Macro-F1 Lead (+0.110)

SerpentGuard leads B1 overall (0.367 vs 0.257) and on post-2022 validation photos unseen by B1 (0.357 vs 0.252), confirming generalizable national superiority across all Indian ecological zones.

### The Honest Scientific Reality

Why does B1 score 0.257? It only knows 62 of India's 184 species, scoring 0 on 122! On the shared 62 species, B1 leads Top-1 (80.1% vs 68.2%) due to global training, highlighting our need to scale from 16% to 100% data in Milestone 4.

| Evaluation Metric | SerpentGuard AI | SnakeCLEF (B1) | Comparison |
|---|---|---|---|
| **Macro-F1 (Full 184)** | **0.367** | 0.257 | Ours leads by +0.110 |
| **Macro-F1 (Post-2022)** | **0.357** | 0.252 | Ours leads by +0.105 |
| **Model File Size** | **10.7 MB (fp16)** | 1,130 MB | Ours is ~1% of B1 |
| **Indian Species** | **184 / 184 (100%)** | 62 / 184 (33.7%) | Full national coverage |
| **Training Photos** | **13,611 (16.4%)** | Full Global Set | Highly data-efficient |
| **Hardware Target** | **Offline Mobile CPU** | Cloud GPU Server | Primary clinic ready |

### 100x Edge Efficiency Advantage

SerpentGuard delivers comparable accuracy at 1% of the model footprint (10.7 MB fp16 vs 1,130 MB server ViT), requiring only 16% of the training data and running fully offline on inexpensive mobile hardware.

**Clinical Reality: A model that only knows 34% of Indian snakes is dangerous in rural practice; 100% taxonomic coverage is non-negotiable.**

CLINICAL SAFETY · DECISION RULES

# Calibrated Clinical Decision Layer & Safety Rules

Enforcing high-recall safety thresholds for urgent medical prioritization

## 94.8% WHO Recall

**Near 95% Target**

95% Confidence Interval: [93.0%, 96.3%]

## 2.1% Lethal Error (O3)

**Tight Guardrail**

Venomous called 'Not Dangerous' capped at ~2.0%

## 0.030 ECE

**Calibrated Uncertainty**

Temperature scaling yields reliable probabilities

### The DR0 Decision Rule

Flags 'Dangerous' if cumulative WHO probability mass ≥ theta_d, top species is venomous, or Viper/Elapid family probability mass ≥ 0.5. Strictly prevents discharging envenomated patients.

### Strict Calibration Protocol

Thresholds fitted strictly on 7,999 photos from unseen photographers. Zero test set leakage ensures completely unbiased clinical estimates across all 184 species without overfitting.

### Acceptable Clinical Trade-off

False alarm rate is 31.1% (down from 42.2%). In emergency medicine, observation is vastly safer than a false discharge of a bitten victim. A false alarm costs observation; a false negative costs a human life.




**Decision Rule Ablation: Evaluated 5 rules (DR0-DR4); DR0 strictly dominated all variants across the safety curve.**

EDGE DEPLOYMENT & MILESTONE 4 ROADMAP

# On-Device Deployment & Milestone 4 Implementation Roadmap

13.58 MB package running at 11.8 FPS on CPU; transitioning to full 85k training & Android APK

## 13.58 MB Total

**Mobile Package Size**

SnakeID fp16 (10.67 MB) + YOLO26n int8 (2.60 MB) + BM25

## 84.9 ms Latency

**End-to-End CPU**

Preprocess 6.8ms + Detector 13.9ms + SnakeID 66ms + Dec 0.35ms

## 11.8 FPS Throughput

**Real-Time Mobile**

Comfortably exceeds the 10 FPS mobile target on single thread

### Quantization & Temporal Voting Highlights

- FP16 SnakeID maintains 100.0% cosine similarity (1.0000) with float32.
- INT8 PTQ detector cuts model size to 2.60 MB with 99.7% box match.
- 4-of-6 video majority vote cuts dangerous flashing from 29.4% to 5.9%.

| Pipeline Stage | Latency | Mobile Budget | Status |
|---|---|---|---|
| **Preprocess (320×320)** | **6.8 ms** | 15.0 ms | Within Target |
| **YOLO26n Detector** | **13.9 ms** | 25.0 ms | Within Target |
| **SnakeID MobileViT-S** | **66.0 ms** | 70.0 ms | Within Target |
| **Decision & BM25** | **0.35 ms** | 5.0 ms | Real-Time |

### M4-1: Full-Dataset Training

Train winning architecture on all 85,067 photos for 40+ epochs on GPU.

### M4-2: Multi-Prototype Clustering

Resolve open-set unknown rejection (current 7.5% vs 70% target) via clustering.

### M4-3: Native Android APK

Embed ONNX Runtime Mobile into Android application for live camera testing.

### M4-4: Expert Blind Review

Conduct planned 200-photo blind clinical review with certified herpetologists.

### M4-5: Field Validation

Test on real mobile video captures in varied Indian rural outdoor environments.

**Group 5 Commitment: Delivering an offline, life-saving AI diagnostic tool for rural Indian healthcare.**